contracts could be used to prevent opioids from being dispensed to misusers while making sure that patients who need it get the medication. In this case, the rules for the governance of drug distribution could be codified into a smart contract shared by all stakeholders in the health care system, including doctors, pharmacies, and governing bodies. This blockchain-based governance approach could easily be expanded to cover the general distribution of any controlled substances and medicines.

1.4.6 Attribution of actions

In many situations, such as research and business workflows, it’s important to know who did what and to whom to attribute the actions taken in a system. Suppose that a patient in a remote rural area with a medical emergency gets transported by ambulance to a major hospital for care. How does the medical insurer decide who gets paid and how much, based on the medical transportation process? The actions taken from the time the call for help was placed to the time when the patient was treated could be recorded in the stakeholders’ ledgers. The payment settlement could be automated through rules governing the rates and services rendered. All this information could be coded in a smart contract.

The smart contract transforms the traditional distributed system into a decentralized system by implementing the rules for compliance, governance, provenance, and information sharing and by recording the necessary details on the blockchain.